MIL-D-13053D

14 July 1969

SUPERSEDING
MIL-D-13053C
3 September 1965

MILITARY SPECIFICATION

DOUGH TROUGHS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers corrosion-resisting steel dough troughs with casters.

1.2 Classification.- Dough troughs shall be of the following sizes, as specified (see 6.2).

Size 3 - 3 feet long

Size 4 - 4 feet long

Size 6 - 6 feet long

Size 8 - 8 feet long

Size 10- 10 feet long

2. APPLICABLE DOCUMENTS

2.1 the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-2316 - Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of.

FSC 7320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Specification A167-63 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19106.)

BAKING INDUSTRY SANITATION STANDARD (BISSC)

No. 2 Dough Troughs

(Copies of Baking Industry Standards Publication may be obtained from the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N.Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample dough trough covered by this specification shall be submitted or made available to the contracting officer

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or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The dough troughs delivered under this specification shall be of the manufacturer's commercially current standard product except for any change necessary to comply with specification requirements.

3.3 Standards.-

3.3.1 BISSC.- Dough troughs shall conform to the applicable requirements of BISSC Standard No. 2.

* 3.3.1.1 BISSC certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dough troughs he proposes to furnish under this specification meet the applicable requirements of BISSC Standard No. 2 as follows:

(1) Display of the BISSC seal on the finished dough trough or a photostatic copy of a current BISSC certificate of authorization for the size dough trough being furnished shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment or

(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the dough troughs are being procured, indicating that the troughs have been tested and conform to BISSC Standard No. 2 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for dough troughs, provided the completed item complies with all provisions of this specification.

* 3.4.1 Corrosion-resisting steel.- Corrosion-resisting steel sheet and strip shall conform to type 302 or 304 of ASTM A 167-63.

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* 3.5 Construction.- The dough trough shall be rectangular in shape. The top of the dough trough shall have a formed rim rolled to 180 degrees on a one inch diameter, minimum; or the rim shall be fully rolled, tightly closed and seal welded against the wall of the trough. The size of the dough troughs shall conform to table I.

TABLE I.- Physical requirements

Size	Length (feet)	Dimensions (minimum) Width, top inside (inches)	Depth, inside (inches)
3	3	21	19
4	4	26	21
6	6	26	21
8	8	26	21
10	10	26	21

3.5.1 Sides and bottom.- The sides and bottom of each trough shall be formed of one sheet of metal and have a 4 inch minimum radius at the intersection of the sides and bottom.

3.5.2 Ends.- The ends shall be made to fit the contour of the sides and bottom and shall be electrically welded to the sides and bottom and shall not crack when tested as specified in 4.5.1. The interior welds shall be filleted to a radius of not less than 1/4 inch.

* 3.5.3 Casters.- Each dough trough shall be furnished with four sanitary type swivel casters bolted to caster shoes. The caster shoes shall be seal welded to the trough body. The swivel assembly shall contain ball or roller bearings in hardened raceways equipped with pressure type lubricating fittings. The wheel bearings shall be ball or roller with pressure type lubricating fittings or lubricated and permanently sealed. The wheels shall be fabricated from a nonmarking material. The wheel size rated capacity for casters applicable to dough trough size shall conform to the requirements of table II.

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TABLE II.- Casters

Trough size	Wheel size (inches)	Wheel width (inches)	Minimum rated capacity (lbs)
3	4	1-7/8	200
4	4	1-7/8	250
6	4	1-7/8	300
8	4	1-7/8	400
10	5	1-7/8	450

The casters when tested as specified in 4.5.1 shall not show any evidence of warping, chipping, binding, looseness or permanent deformation.

3.6 Finish.- The finish on all dough troughs shall conform to BISSC Standard No. 2.

* 3.7 Marking for identification.- Each dough trough shall be marked with the manufacturer's name, trade name, or trademark readily identifiable with said manufacturer. Marking shall be applied in accordance with MIL-STD-130.

3.8 Workmanship.-

3.8.1 Fabrication.- The finished dough trough shall be clean, well made, and all sharp burrs, slivers or splinters shall be removed.

3.8.2 Welding.- All surface of parts to be welded shall be clean. Welding shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All welds shall be continuous. All scale or flux deposits shall be removed from the finished welds.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects listed in table III, dimensions specified herein, and tested as specified in 4.5.1. Failure to pass any tests shall be cause for rejection of the preproduction sample. The presence of any defects listed in table III or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.3.2 End item inspection.- A lot shall consist of all dough troughs of the same size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated dough trough.

4.3.2.1 Visual examination.- Examination shall be made of the end item for the defects listed in table III. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 for major defects, and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III.- Classification of defects

Examine	Defect	Classification Major	Minor
Construction and workmanship (applicable to all components and assemblies)	Sides and bottom not formed of one piece of metal.	X	
	Rim not fully rolled, tightly closed or not constructed so as to be readily cleaned.		X

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TABLE III.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction and workmanship (applicable to all components and assemblies) (cont'd)	Any sharp burrs, slivers or exposed splinters.	X	
	Any component malformed or damaged.	X	
Welding	Not continuous	X	
	Scale or flux deposits not removed, food contact area.	X	
	Scale or flux deposits not removed, non-food-contact area.		X
	Interior welds not filleted to a radius of 1/4 inch.	X	
	Cracked or fractured.	X	
	Burn holes.	X	
Casters	Not type, specified.	X	
	Caster shoe not welded to trough body.	X	
	Not equipped with pressure type lubrication fittings (when applicable).	X	
	Casters do not swivel freely.	X	
Identification marking	Incomplete, incorrect, or illegible		X
	Marking not on nameplate, or nameplate not securely affixed in conspicuous place.		X

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4.3.2.2 Dimensional examination.- Inspection shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be level S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 Testing of the end item.- The end item shall be tested as specified in 4.5.1. The inspection level shall be S-2 and the AQL shall be 2.5 defects, expressed in terms of defects per hundred units.

4.3.3 Examination for preparation for delivery.- Examination shall be made for preservation, packaging, packing and marking in accordance with MIL-B-2316.

* 4.4 Code and standard compliance.- The supplier shall have available for the Government representative proof of compliance with the requirements of 3.3.

4.5 Test.-

4.5.1 The dough troughs, as applicable, shall be loaded with an evenly distributed load of: Size 3, 600 pounds; Size 4, 800 pounds; Size 6, 1100 pounds; Size 8, 1500 pounds; Size 10, 1800 pounds, and shall be rolled up and over an obstacle 4 times. The obstacle shall have vertical faces, a minimum of 1/8 inch high, shall be at least 12 inches wide, and shall be long enough to allow both leading casters to strike the obstacle at the same time. The dough trough shall be examined to determine compliance with the welding requirements specified in 3.5.2 and the caster requirements specified in 3.5.3. Any cracks in the welds shall be cause for failure of the test.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C.- Preservation, packaging, packing, and marking for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2)

6. NOTES

6.1 Intended use.- The size 3 dough trough covered by this specification is used primarily in the Department of Defense standardized

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1000 and 1500 man dining facilities. Sizes 4, 6, 8 and 10 dough troughs covered by this specification are used primarily in fixed bakeries and pastry kitchens.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of dough trough required (see 1.2).
(c) First article.- When a preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of preservation and packaging and packing (see 5.1).

6.3 The margins of this specification have been marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) were made from previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0256

Review activities:

Army - MD
Navy - MC, MS, SA, YD
Air Force - 84

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Natick, Massachusetts 01760

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Dough Troughs MIL-D-13053D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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